(ii) The cosine function, cos x

$$\text{Let } y = \cos x = \sin\left(x+\frac{\pi}{2}\right).$$

$$\text{Then, } \frac{dy}{dx} = \frac{d}{dx}\sin\left(x+\frac{\pi}{2}\right)$$

$$\text{Let } u = x+\frac{\pi}{2}$$

$$\frac{du}{dx} = 1+0=1$$

$$\text{Therefore, } \frac{dy}{dx} = \frac{d}{dx}(\sin u) = \frac{d}{dx}(\sin u)\frac{du}{dx} \text{ by Chain rule}$$

$$= \cos u \times 1 = \cos u = \cos\left(x+\frac{\pi}{2}\right) = -\sin x \ .$$

That is, $\boxed{\frac{d}{dx}(\cos x) = -\sin x}$.

(iii) The tangent function, tan x

$$\text{Let } y = f(x) = \tan x \ .$$

$$= \frac{\sin x}{\cos x}$$

$$\text{Therefore, } \frac{d}{dx}(\tan x) = \frac{d}{dx}\left(\frac{\sin x}{\cos x}\right)$$

$$= \frac{\cos x \frac{d}{dx}(\sin x) - \sin x \frac{d}{dx}(\cos x)}{\cos^2 x} \quad \text{(by quotient rule)}$$

$$= \frac{\cos x(\cos x) - \sin x(-\sin x)}{\cos^2 x}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$$

$$= \frac{1}{\cos^2 x} \ .$$

That is, $\boxed{\frac{d}{dx}(\tan x) = \sec^2 x}$.

(iv) The secant function, sec x

$$\text{Let } y = \sec x = \frac{1}{\cos x} = (\cos x)^{-1} \ .$$

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